UNIVERSITI PENDIDIKAN SULTAN IDRIS
SULTAN IDRIS EDUCATION UNIVERSITY

UNIVERSITI PENDIDIKAN SULTAN IDRIS

FINAL EXAMINATION
SEMESTER 1 SESSION 2015/2016

CODE : MES3103 **COURSE : SOFTWARE VERIFICATION AND VALIDATION**

DATE : 28 DEC 2015 **DURATION : 2 HOUR 30 MINUTES**

INSTRUCTIONS

1. This question paper consists of **TWO (2)** sections only, namely **Section A** and **Section B**.
2. **Section A** : Answer **ALL** questions in the answer booklet provided.
3. **Section B** : Answer **ALL** questions in the answer booklet provided.
4. This question paper will be collected at the end of the examination.

This question paper consists of 6 printed pages including front page

PROGRAMME : _____

YEAR : _____ CLASS GROUP : _____

REGISTRATION NO : _____ | | | | | | |

IDENTITY CARD NO : | | | | | | | | | | | | |

LECTURER : DR. NOOR AZIDA BINTI SAHABUDIN

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SECTION A: Multiple Choice Questions (10 MARKS)

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Instructions: Answer **all** questions in the answer booklet.

1. Which of the following statement is **TRUE** about verification and validation based on ISO 9000?

   A. Verification is a confirmation by clients and through the provision of objective evidence that specified requirements have be fulfilled; validation is a confirmation by client and through provision of objective evidence that the requirements for a specific intended use or application have be fulfilled.
   B. Verification is a confirmation by clients and through the provision of objective evidence that specified requirements have be fulfilled; validation is a confirmation by examination and through provision of objective evidence that the requirements for a specific intended use or application have be fulfilled.
   C. Verification is a confirmation by examination and through provision of objective evidence that the requirements for a specific intended use or application have be fulfilled; validation is a confirmation by examination and through the provision of objective evidence that specified requirements have be fulfilled.
   D. Verification is a confirmation by examination and through the provision of objective evidence that specified requirements have be fulfilled; validation is a confirmation by examination and through provision of objective evidence that the requirements for a specific intended use or application have be fulfilled.

2. In pre-project components, contract review includes the following activities **EXCEPT**

   A. clarification of the customer's requirements
   B. evaluation of the product's capacity to fulfill his obligations
   C. review of the project's schedule and resource requirement estimates
   D. evaluation of the professional staff's capacity to carry out the proposed project

3. The main issues treated in the project's quality plan are:

   A. quality goal, expressed in the appropriate measureable terms, criteria for starting and ending each project stage, lists of reviews, tests, and other scheduled verification and validation activities
   B. quality plan, expressed in the appropriate measureable terms, criteria for starting and ending each project stage, lists of reviews, tests, and other scheduled verification and validation activities
   C. quality goal, expressed in the appropriate measureable terms, criteria for starting and ending each project stage, lists of product, tests, and other scheduled verification and validation activities
   D. quality plan, expressed in the appropriate measureable terms, criteria for starting and ending each project stage, lists of product, tests, and other scheduled verification and validation activities

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4. Which of the following statement is **NOT TRUE** about the importance of quality metrics?

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   A. To determine the quality and progress of testing.
   B. To estimate the time needed to produce software.
   C. To calculate how much more time is required for the release.
   D. To help in deciding the scope of the product that can be released by considering the defect density across modules.

5. To analyzing metrics, the following Ishikawa's 7 "old" tools can be used

   A. Checklist, Pareto Diagram, Histogram, Scatter diagram, Run Chart, Control Chart, Microsoft excel
   B. Checklist, Pareto Diagram, Histogram, Scatter diagram, Run chart, Control Chart, Cause-and-effect diagram
   C. Checklist, Pareto Diagram, Histogram, Scatter diagram, Control Chart, Cause-and-effect diagram, Microsoft excel
   D. Checklist, Pareto Diagram, Histogram, Scatter diagram, Run chart, Control Chart, Cause-and-effect diagram, Microsoft excel

6. Which of the following statement is **NOT TRUE** about formal and informal review?

   A. It is useful to combine informal and formal reviews
   B. An informal review is a verbal review requested by a client of a work product to improve that product
   C. Informal reviews help identify simple defects. Formal reviews can identify complex defects that might be missed by simple desk checks
   D. A formal review is a written review conducted by a team leader or a moderator for the purpose of identifying, documenting, and fixing defects in a work product

7. Walkthrough includes the following process **EXCEPT**

   A. identify resources
   B. examine technical alternatives
   C. evaluate checklists, standards, and rules
   D. discussion of deliverable style and format

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8. Which of the following statement is **NOT TRUE** about white box testing and black box testing?

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   A. White box testing is the software testing method in which internal structure is being known to tester who is going to test the software.
   B. Black box testing is the Software testing method which is used to test the software without knowing the internal structure of code or program.
   C. Black box testing is primarily concentrated on the testing of program code of the system under test like code structure, branches, conditions, loops etc.
   D. White box testing is primarily concentrated on the testing of program code of the system under test like code structure, branches, conditions, loops etc.

9. Heuristic Evaluation Check List includes the following tasks **EXCEPT:**

   A. error prevention
   B. quality of product
   C. recognition rather than recall
   D. flexibility and efficiency of use

10. Which of the following statement is **TRUE** about Test Maturity Model Integration (TMMi)?

   A. TMMi uses the concept of maturity levels for process design and development
   B. The TMMi has been developed to support organizations with evaluating and improving their product
   C. TMMi is a detailed model to test the product and it is positioned as being complementary to the CMMI
   D. TMMi framework has been developed by the TMMi Foundation as a guideline and reference framework for test process improvement and is positioned as a complementary model to the CMMI Version 1.2

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SECTION B: Short Answer Questions (60 MARKS)

Instructions: Answer **all** questions in the answer booklet provided.

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1. McCalls's factor model classifies all software requirements into 11 software quality factors. These factors are grouped into 3 categories (product operation, product revision and product transition). Identify the factors of each category.

   [10 marks]

2. Software quality assurance (SQA) is a process that ensures that developed software meets and complies with defined or standardized quality specifications. SQA is an on-going process within the software development life cycle (SDLC) that routinely checks the developed software to ensure it meets desired quality measures and it involved six classes of components. Can you distinguish between **six (6)** classes of SQA system components?

   [12 marks]

3. One of the biggest American market makers for stocks struggled to stay afloat after a software bug triggered a $440 million loss in just 30 minutes. The firm's shares lost 75 percent in two days after the faulty software flooded the market with unintended trades. One of Knight's trading algorithms reportedly started pushing erratic trades through on nearly 150 different stocks, sending them into spasms. List **seven (7)** reasons, why this event could have happened?

   [7 marks]

4. Software metric is a combination of two or more measures used to compare software processes, projects and products. Set of metrics derived from the data collected from past projects at the organization level.

   a) Based on your knowledge, what makes a good software metrics?

   [5 marks]

   b) Metris can give benefits to the finance, process, customer and people. Can you write a brief outline about the benefits of metrics in those areas.

   [10 marks]

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5. A review is any activity where reviewers examine a work product to provide feedback. Reviews can reveal defects early, when the defects are less costly to fix.

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a) There are two types of review which are formal review and informal review. Make a comparison about this review.

[4 marks]

b) Ahmad is a group leader of a project in development of a software product. He and his team members (moderator, programmer, designer, reader and reviewer, recorder and producer) are decide to review the project before submit to the client. Compose step-by-step, what should Ahmad and his team members need to do.

[12 marks]

End of Question

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